

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1899

Mount View 138 kV
August 26, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1899. The Interconnection Customer proposes a 144.6 MW battery energy storage facility interconnecting at the Mount View 138 kV substation in the Virginia area. The request was received on October 07, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 480 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,416,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Mount View - Hartman Creek 138.0 kV
Contingency: Lowmoor - Hinton Ap 345.0 kV line
Loading: 114.4% (Rating: 328 MVA)
DFAX: 80.6%
Pre-project loading: 78.8%
2. Facility: Parke 138/1 kV TX
Contingency: New Hope Ap - Barr Lake 345.0 kV Line
Loading: 101.2% (Rating: 482 MVA)
DFAX: 10.0%
Pre-project loading: 98.1%
3. Facility: Parke - Taloga 138.0 kV
Contingency: Barr Lake - New Hope Ap 345.0 kV
Loading: 101.2% (Rating: 482 MVA)
DFAX: 10.0%
Pre-project loading: 98.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mount View - Hartman Crk 138.0 kV	Rebuild line with higher-capacity conductor	\$2,744,000	41% (\$1,119,000)
Parke 138/1 kV Transformer	Replace transformer with higher-rated unit	\$5,409,000	4% (\$190,000)
Parke - Taloga 138.0 kV	Reconductor line	\$3,054,000	4% (\$107,000)
	Total allocated to project		\$1,416,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.